ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (pp. 3-10)
- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
- Minimum, Maximum, Mixed Phase Filters (PP. 24-28)
- All-Pass filters (pp. 29-31)
- Tutorial questions (pp. 32-38)

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Term 3, 2021

Digital Filter Fundamentals:

Introduction

- A digital filter is a discrete-time system that is designed to remove some frequency components or modify some characteristic of a signal, but often the two terms (systems or filters)are used interchangeably.
- There are two types of digital filters:
  (a) FIR (Finite Impulse Response) filters or Non-recursive filters (no feedback path)
  (b) IIR (Infinite Impulse Response) filters or Recursive filters (at least one feedback path present)
- A linear time invariant (LTI) discrete-time system described by the following equation is commonly called a digital filter:

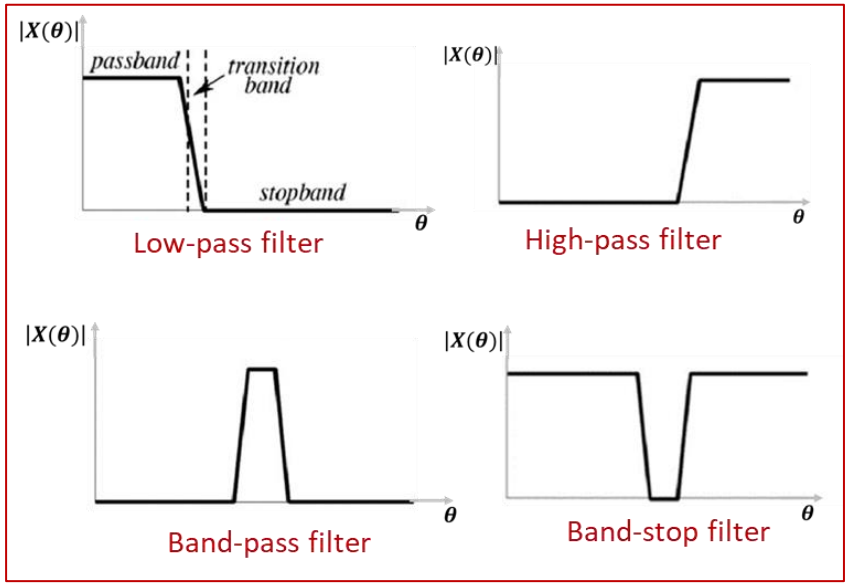
$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

Feed forward Feedback

where $x[n]$ is the input signal, $y[n]$ is the output signal. a_0, a_1 , a_2, ..., a_M; b_1, b_2, b_3, ..., b_L are constants (filter coefficients). These coefficients determine the characteristics of the filter.

- **When $b_k = 0$, the filter is an FIR filter ; when $b_k \neq 0$, the filter is an IIR filter.**





Four Common Digital Filters

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Digital Filter Fundamentals: FIR filter

FIR filter

- Any filter that has an impulse response of finite duration is called Finite Impulse Response (FIR) filter.
- If $b_k = 0$, then the calculation of *y[n]* does not require the use of previously calculated samples of the output (see the equation below):

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

$$\therefore \quad y[n] \quad = \quad \sum_{k=0}^{M} a_k x[n-k]$$

$$= \quad a_0 x[n] + a_1 x[n-1] + \cdots + a_M x[n-k]$$

- This is recognised as a convolution sum:

$$y[n] = \sum_{k=0}^{M} h[k] x[n-k]$$

- Therefore, the impulse response is identical to the coefficients, that is,

$$h[n] = \begin{cases} a_n & 0 \le n \le M \\ 0 & otherwise \end{cases}$$

Example 1:

- The following is a (non-recursive) second order FIR filter:

$$y[n] \quad = a_0 x[n] + a_1 x[n-1] + a_2 x[n-2]$$

$$H(z) = \frac{Y(z)}{X(z)} = a_0 + a_1 z^{-1} + a_2 z^{-2}$$

FIR filter properties

A property of the FIR filter is that it will always be stable:

- **Stability** requires that there should be no poles outside the unit circle. This condition is automatically satisfied since there are no poles at all outside the origin. (In fact, all poles are located at the origin.)
- Another property of FIR filter is that we can make filters with exactly **linear phase** characteristics.
- The ability to have an exactly linear phase response is one of the most important properties of a LTI system (filter).

Digital Filter Fundamentals: FIR filter – Phase Delay & Group Delay

Phase Delay or Group Delay

- When a signal passes through a filter, it is modified in amplitude and/or phase.
- The nature and extent of the modification of the signal is dependent on the amplitude and phase characteristics of the filter.
- The **phase delay** or **group delay** of the filter provides a useful measure of how the filter modified the phase characteristic of the signal.



Phase Delay

- If we consider a signal that consists of several frequency components (eg. speech waveform) the *phase delay* of the filter is the amount of time delay each frequency component of the signal suffers in going through the filter.

$$Phase_delay(T_p) = -\frac{\emptyset(\theta)}{\theta}$$

T_p = the negative of the phase angle divided by frequency

Group Delay

- The *group delay* on the other hand is the average time delay the composite signal suffers at each frequency as it passes from the input to the output of the filter.

$$Group_delay(T_g) = -\frac{d\emptyset(\theta)}{d\theta}$$

T_g = the negative of the derivative of the phase with respect to frequency

Digital Filter Fundamentals: Linear phase FIR filter

Linear Phase Filter

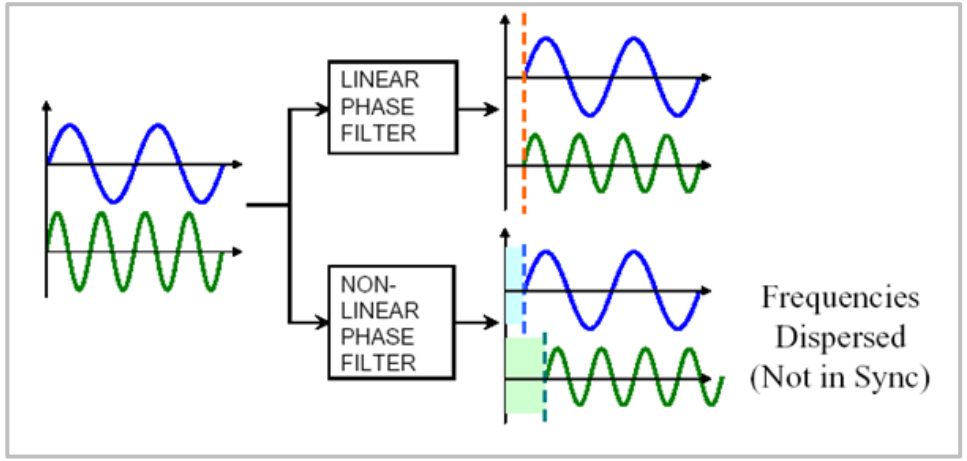
- A *constant group delay* means that signal components at different frequencies receive the same delay in the filter.
- A *linear phase filter* gives same time delay to all frequency components of the input signal. A filter with a nonlinear phase characteristic will cause a phase distortion in the signal that passes through it.
- This is because the frequency components in the signal will each be delayed by an amount not proportional to frequency, thereby altering their harmonic relationship. Such a distortion is undesirable in many applications, for example music, video etc.
- A filter is said to have a **linear phase response** if its phase response satisfies one of the following relationships:

$$\emptyset(\theta) = -a\theta$$

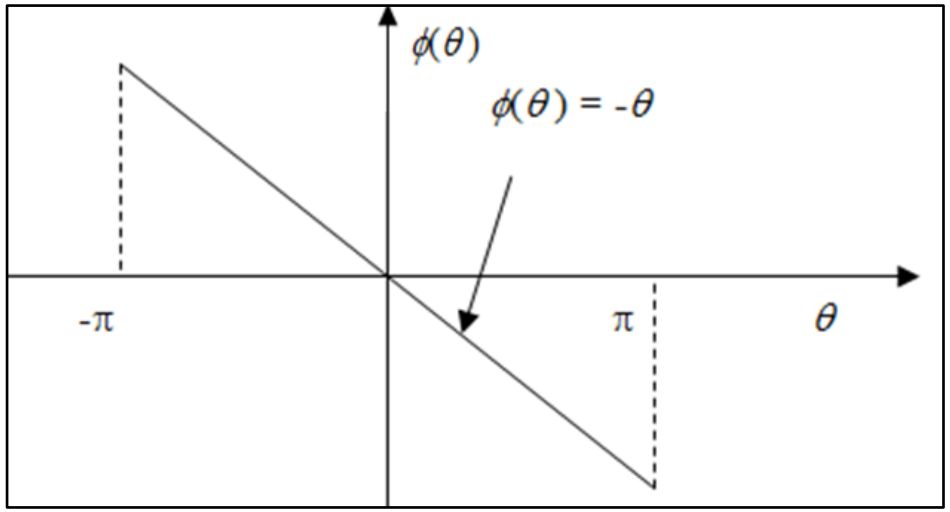
$$\emptyset(\theta) = b - a\theta$$

where '*a*' and '*b*' are constants.

Comparing a linear phase filter with a non-linear phase filter



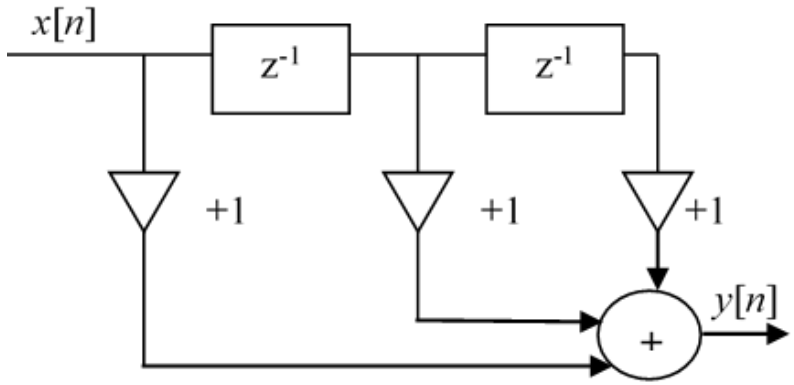
Phase Response of a linear Phase filter



Phase varies linearly with frequency

Digital Filter Fundamentals: Linear phase FIR filter

Example 2: Two filter structures are shown below. Show that both filters have linear phase.



$$y[n] = x[n] + x[n-1] + x[n-2]$$

$$H(z) = 1 + z^{-1} + z^{-2}$$

$$H(\theta) = 1 + e^{-j\theta} + e^{-j2\theta}$$

$$= e^{-j\theta}\left(e^{j\theta} + e^{-j\theta} + 1\right)$$

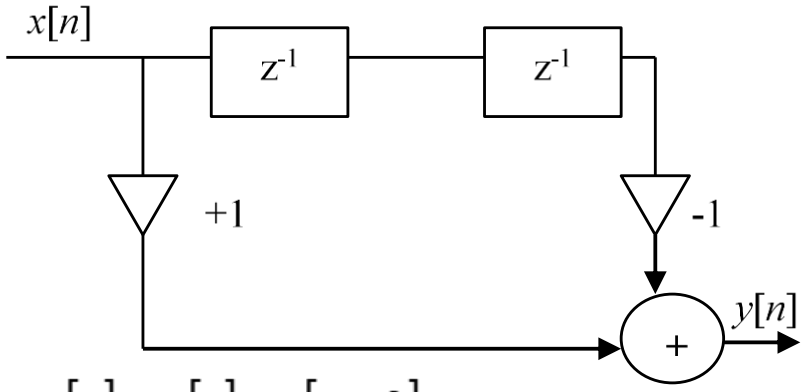
$$= e^{-j\theta}(1 + 2\cos\theta)$$

Let $B(\theta) = 1 + 2cos\theta$

Phase: $\phi(\theta) = -\theta \quad B(\theta) \geq 0$

$\phi(\theta) = -\theta \ \pm\pi \quad B(\theta) < 0$

- If $\emptyset(\theta) = -\theta\ (Linear\ Phase)$ the impulse response is always even symmetric about its midpoint.



$$y[n] = x[n] - x[n-2]$$

$$H(z) = 1 - z^{-2}$$

$$H(\theta) = 1 - e^{-j2\theta}$$

$$= 2je^{-j\theta}\left(\frac{e^{j\theta} - e^{-j\theta}}{2j}\right)$$

$$= 2e^{j\frac{\pi}{2}}e^{-j\theta}\sin\theta$$

$$= e^{j\left(\frac{\pi}{2}-\theta\right)}(2\sin\theta)$$

Let $B(\theta) = 2sin\theta$

Phase: $\phi(\theta) = \pi/2 - \theta \quad B(\theta) \geq 0$

$\phi(\theta) = (\pi/2 - \theta) \pm\pi \quad B(\theta) < 0$

- If $\emptyset(\theta) = \pi/2 - \theta$, the impulse response is always odd symmetric about its midpoint

The frequency response of linear phase filter may be written as

$$H(\theta) = A(\theta)\, e^{j\alpha\theta}$$

(for even symmetric h[n])

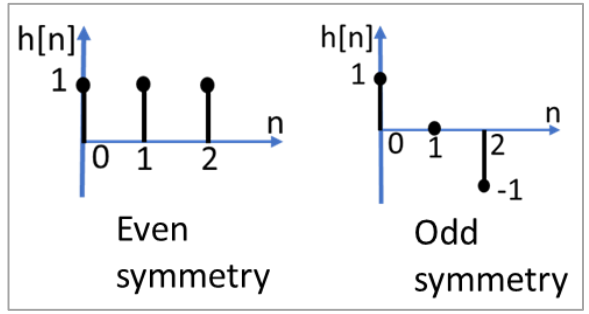
or $H(\theta) = je^{j\alpha\theta}A(\theta)$

$$= e^{j\left(\alpha\theta+\frac{\pi}{2}\right)}A(\theta)$$

(for odd symmetric h[n])

where $A(\theta)$ is real.

For a linear phase FIR filter its coefficients are symmetrical about its midpoint

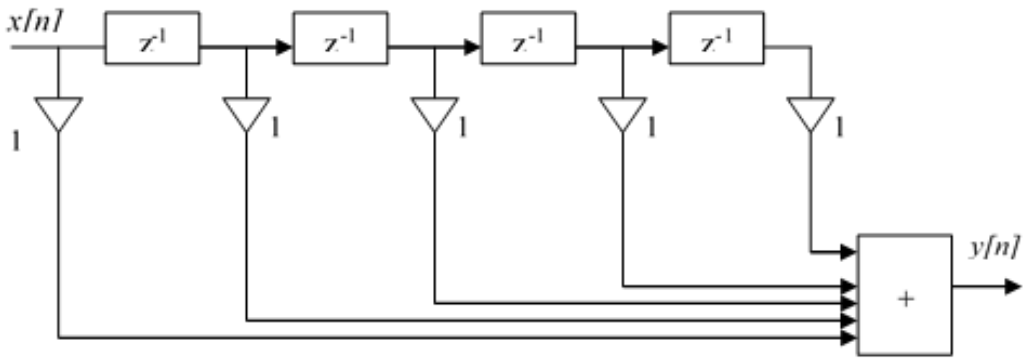


Digital Filter Fundamentals: Linear phase FIR filter

Example 3:

a) Show that the filter structure shown on the right has a linear phase characteristic equation given by: $\phi(\theta) = -2\theta$.

b) Show for the above liner phase filter , both the phase delay and the group delay are constant.



Answer:

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

Let N = number of coefficients of the liner phase filter

$$H(z) = \frac{1 - z^{-N}}{1 - z^{-1}} \Rightarrow H(\theta) = \frac{1 - e^{-j\theta N}}{1 - e^{-j\theta}}$$

$$H(\theta) = \frac{e^{-j\frac{\theta N}{2}}}{e^{-j\frac{\theta}{2}}} \frac{[e^{j\frac{\theta N}{2}} - e^{-j\frac{\theta N}{2}}]}{[e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}]} = e^{-j\frac{N-1}{2}\theta} \frac{\sin\left(\frac{N\theta}{2}\right)}{\sin\frac{\theta}{2}}$$

$$N = 5 \quad \Rightarrow \quad H(\theta) = e^{-j2\theta}\left\{\frac{\sin\frac{5\theta}{2}}{\sin\frac{\theta}{2}}\right\} = B(\theta)$$

$\therefore \phi(\theta) = -2\theta \qquad B(\theta) \geq 0$

$\phi(\theta) = -2\theta \ \pm\pi \quad B(\theta) < 0$

$$Phase_delay(T_p) = -\frac{\varphi(\theta)}{\theta} = -\frac{-2\theta}{\theta}$$

$$= 2\ (constant)$$

$$Group_delay(T_g) = -\frac{d\varphi(\theta)}{d\theta} = -\frac{d[-2\theta)]}{d\theta}$$

$$= 2\ (constant)$$

Digital Filter Fundamentals: IIR filter

IIR filter

- Every recursive (Infinite Impulse Response) digital filter must contain at least one closed loop. Each closed loop contains at least one delay element.

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

- For recursive digital filters $b_k \neq 0$. For example, $a_k = 0$ for $k \geq 1$ and & also $b_k = 0$ for $k \geq 1$.

$$\therefore \quad y[n] = a_0 x[n] - b_1 y[n-1]$$

$$H(z) = \frac{a_0}{1+b_1 z^{-1}} \qquad \text{[IIR filter]}$$

- A recursive filter is an infinite impulse response filter (IIR).

Second Order Filters

Transfer function	Description
$H_1(z) = a_0 + a_1 z^{-1} + a_2 z^{-2}$	all-zero second order FIR filter (excluding poles at origin)
$H_2(z) = \dfrac{1}{1+b_1 z^{-1} + b_2 z^{-2}}$	all-pole second order IIR filter (excluding zeros at origin)
$H_3(z) = \dfrac{1+a_1 z^{-1} + a_2 z^{-2}}{1+b_1 z^{-1} + b_2 z^{-2}}$	pole-zero second order IIR filter

Note: Poles and zeros can be real or imaginary

Digital Filter Fundamentals: IIR filter – DC gain and 3 dB bandwidth

Example 4: DC gain calculation

- Consider the 1st order IIR filter

$$H(z) = \frac{1}{1 - az^{-1}} \quad 0 < a < 1$$

The difference equation is:

$$y[n] = x[n] + a\,y[n-1].$$

- The DC gain of $H(z)$ can be obtained by substituting $\theta = 0$.

$$H(\theta)|_{\theta=0} = \frac{1}{1 - ae^{-j(0)}} = \frac{1}{1-a}$$

- If DC gain is undesirable, introduce a constant gain factor of (1-a), so that $H(z)$ becomes

$$H(z) = \frac{1-a}{1-az^{-1}} \quad \text{[DC gain =1]}$$

$$\therefore\ y[n] = (1-a)\,x[n] + a\,y[n-1]$$

Example 5: 3dB bandwidth calculation

- Consider a lowpass filter $y[n] = ay[n-1] + bx[n], 0 < a < 1$

(i) Determine '*b*' so that $|H(0)| = 1$.
(ii) Determine the 3dB bandwidth for the normalised filter in part (i).

Answer:

$$Y(z) = a\,Y(z)z^{-1} + b\,X(z) \Rightarrow H(z) = \frac{b}{1-az^{-1}} \Rightarrow H(\theta) = \frac{b}{1-ae^{-j\theta}}$$

we have $|H(0)| = 1;\ H(0) = \frac{b}{1-a} = 1 \Rightarrow b = 1 - a$

(ii) $H(\theta) = \frac{1-a}{1-ae^{-j\theta}} = \frac{1-a}{(1-a\cos\theta)+ja\sin\theta};\ |H(\theta)| = \frac{1-a}{\sqrt{1+a^2-2a\cos\theta}}$

Second Method: $|H(\theta)|^2 = H(\theta)\cdot H^*(\theta)$ {$H^*(\theta)$ is the complex conjugate of $H(\theta)$}

$$\left\{|H(\theta)|^2 = \frac{1-a}{1-ae^{-j\theta}}\cdot\frac{1-a}{1-ae^{j\theta}} = \frac{(1-a)^2}{1-2a\cos\theta+a^2}\right\}$$

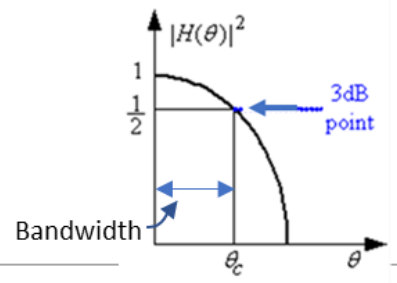
θ_c is the cut-off frequency at which:

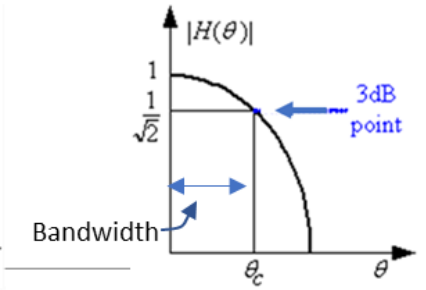
$$|H(\theta)|^2_{\theta=\theta_c} = \frac{1}{2}|H(0)|^2 \quad \textbf{or} \quad |H(\theta)|_{\theta=\theta_c} = \frac{1}{\sqrt{2}}|H(0)|$$

We have $|H(0)|^2 = 1$ *when* $\theta = 0$;

$$\frac{1}{2} = \frac{(1-a)^2}{1+a^2-2a\cos\theta_c}$$

$$\theta_c = \cos^{-1}\left[\frac{-a^2+4a-1}{2a}\right]$$





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Digital Filter Fundamentals: Digital filter Structures

Digital Filter Structures

- It is common for a digital filter to be implemented by combining simple digital filter structures (e.g. Direct forms I & II) to form a larger composite filter.
- There are two common ways of combining simple structures, either in **cascade** or in **parallel**.

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] - \sum_{k=1}^{L} b_k y[n-k]$$

$$Y(z) = \sum_{k=0}^{M} a_k X(z) z^{-k} - \sum_{k=1}^{L} b_k Y(z) z^{-k}$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} a_k z^{-k}}{1 + \sum_{k=1}^{L} b_k z^{-k}}$$

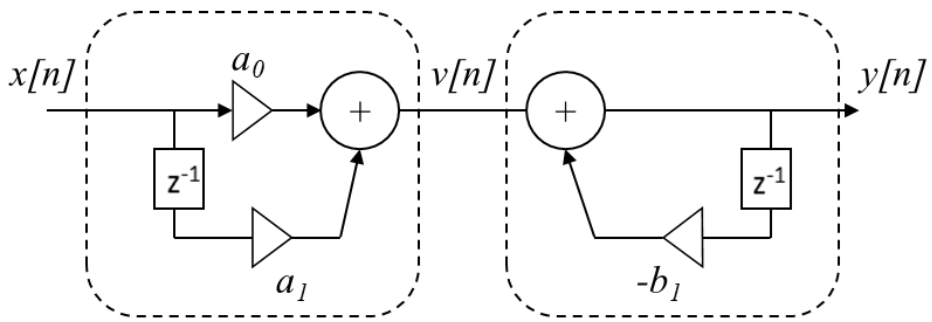
- **Direct form I:**

$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$

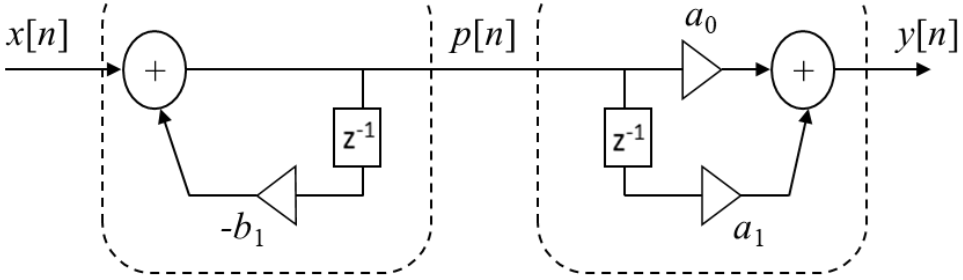
- **Direct form II:**

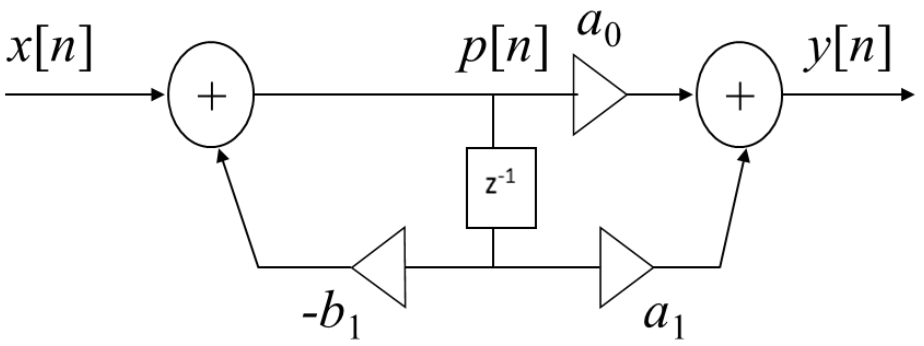
$$H(z) = \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$

Direct form I



Direct form II





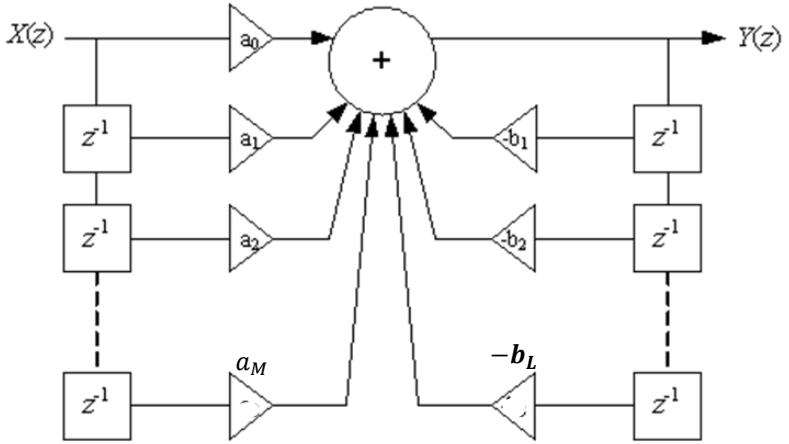
Canonic form

Digital Filter Fundamentals: Digital filter Structures

Direct Form I:

- **Direct form I:**

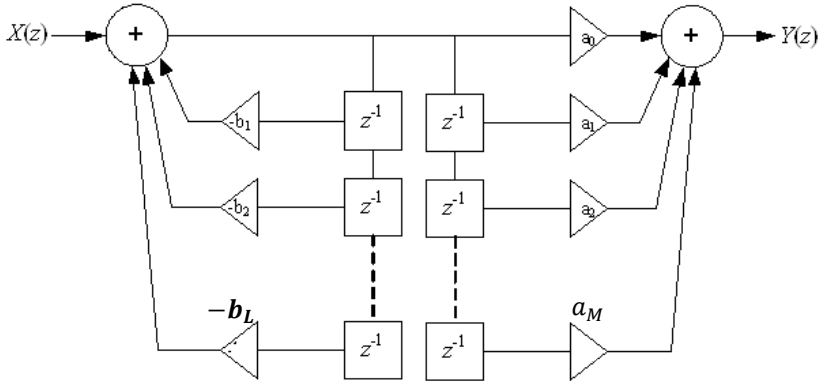
$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$



Direct Form II:

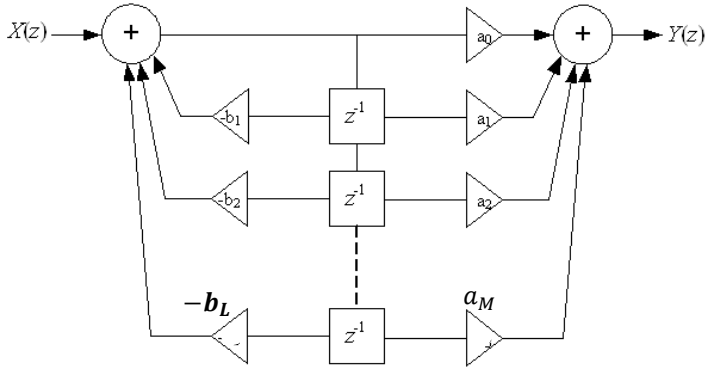
- **Direct form II:**

$$H(z) = \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$

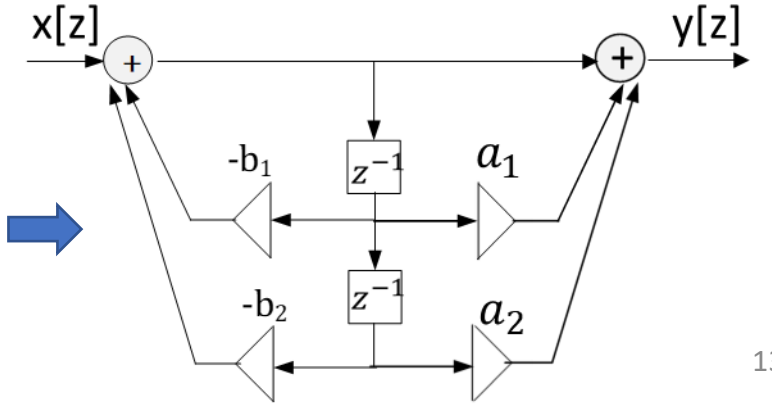


Canonic Form:

- In the case when *L* = *M*, we have canonic form realisation.



A discrete-time filter is said to be canonic if it contains the minimum numbers of delay elements necessary to realise the associated frequency response.



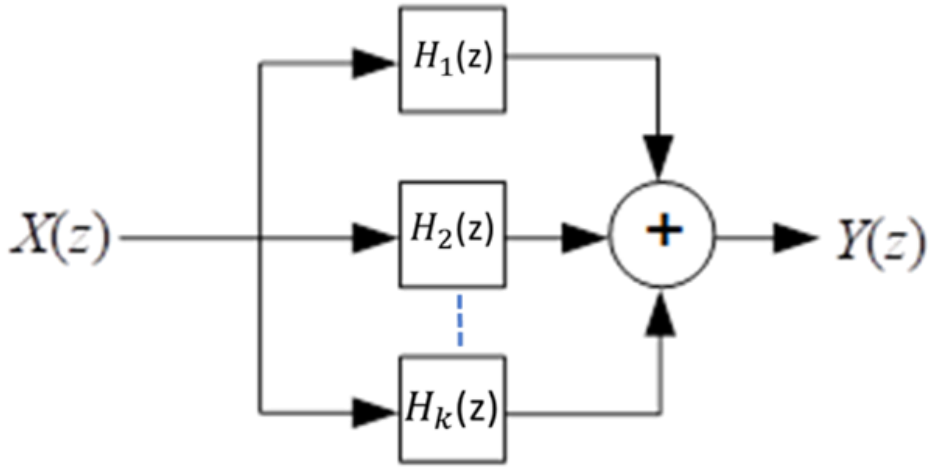
Digital Filter Fundamentals: IIR filter – Parallel Structure

Parallel Structure

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$= \underbrace{\sum_{i=1}^{k} H_i(z) \quad = \quad H_1(z) + H_2(z) + H_3(z) + \ldots + H_k(z)}_{parallel_structure}$$

Note: Use partial fraction to obtain $H_i(z)$



Parallel Structure

Example 7:

- Draw a parallel realisation of a third order system $H(z)$ as given below: $H(z) = \frac{z^3 - z^2 - 5z - 3}{5z(z^2 - 4)}$

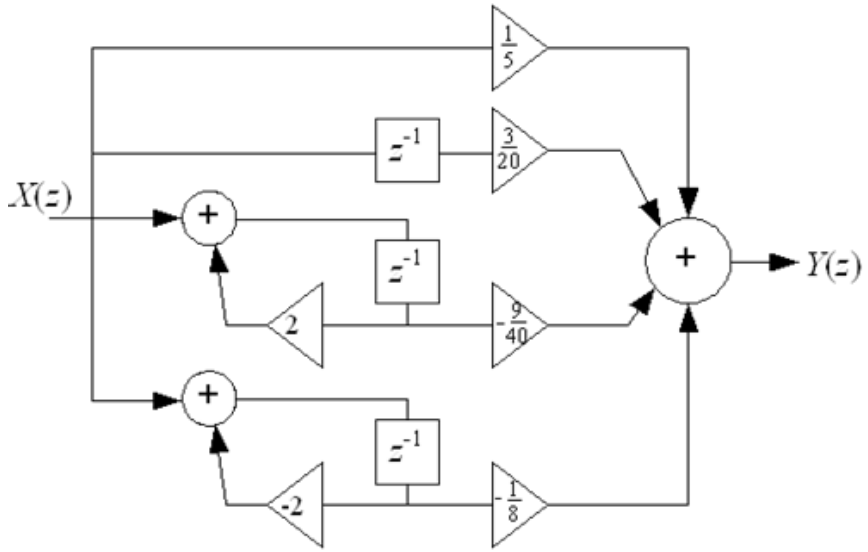
Answer:

By using a partial fraction decomposition, $H(z)$ becomes:

$$H(z) = A + \frac{B}{z} + \frac{C}{z-2} + \frac{D}{z+2}$$

$$H(z) = \frac{1}{5} + \frac{3}{20z} - \frac{9}{40(z-2)} - \frac{1}{8(z+2)}$$

$$H(z) = \frac{1}{5} + \frac{3}{20} z^{-1} - \frac{9z^{-1}}{40(1-2z^{-1})} - \frac{z^{-1}}{8(1+2z^{-1})}$$



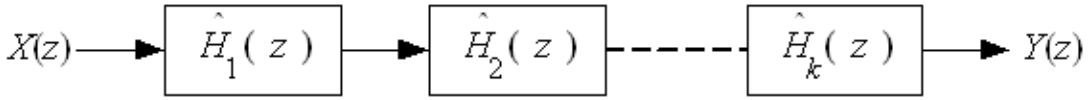
Digital Filter Fundamentals: IIR filter – Cascade Structure

Cascade Structure

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$H(z) = \prod_{i=1}^{k} \hat{H}_i(z) = \underbrace{\hat{H}_1(z) \cdot \hat{H}_2(z) \cdot \hat{H}_3(z) \ldots \hat{H}_k(z)}_{cascade\ structure}$$

Note: H(z) now becomes the product of lower order transfer functions $\hat{H}_1(z)$, $\hat{H}_2(z)$ *etc* i.e. 1st or 2nd order sections. The cascade structure is the most popular form in digital filter implementation.



Cascade Structure

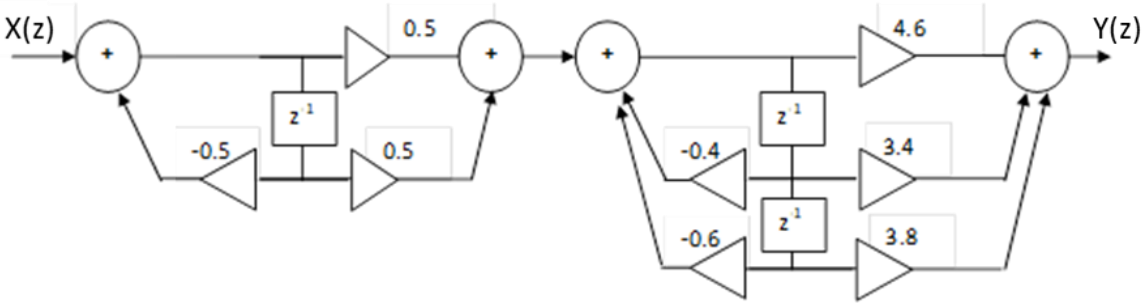
Example 8:

- A cascade realisation of a third-order filter is given by

$$H(z) = \frac{23 + 40z^{-1} + 36z^{-2} + 19z^{-3}}{10 + 9z^{-1} + 8z^{-2} + 3z^{-3}} = \frac{(1 + z^{-1})}{(2 + z^{-1})} \frac{23 + 17z^{-1} + 19z^{-2}}{5 + 2z^{-1} + 3z^{-2}}$$

$$= \frac{(1 + z^{-1})}{2(1 + 0.5z^{-1})} \frac{23 + 17z^{-1} + 19z^{-2}}{5(1 + 0.4z^{-1} + 0.6z^{-2})} = \left(\frac{(0.5 + 0.5z^{-1})}{1 + 0.5z^{-1}}\right)\left(\frac{4.6 + 3.4z^{-1} + 3.8z^{-2}}{1 + 0.4z^{-1} + 0.6z^{-2}}\right)$$

1st order 2nd order



Example 9:

- Draw a cascade realisation of the filter H(z):

$$H(z) = \frac{1}{(1 - az^{-1})^3}$$

Note: No parallel structure exists for H(z), because partial fraction expansion cannot be performed



ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (pp. 3-10)
- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
- Minimum, Maximum, Mixed Phase Filters (PP. 24-28)
- All-Pass filters (pp. 29-31)
- Tutorial questions (pp. 32-38)

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Term 3, 2021

Digital Filter Fundamentals: Magnitude and Phase Responses

Phase Response

- We can show that the magnitude response is an **even** function of frequency.
- The phase response is an **odd** function of frequency.
- For example, let us consider the following linear-phase form of the transfer function:

$$H(\theta) = e^{-jk\theta} B(\theta) \quad k > 0$$

The real-valued function of $B(\theta)$ that can take positive and negative values.

$$H(\theta) = B(\theta)\cos(-k\theta) + B(\theta)j\sin(-k\theta)$$

$$H(\theta) = B(\theta)\cos(k\theta) - jB(\theta)\sin(k\theta)$$

Let the phase angle be ϕ:

$$\tan\phi = -\frac{B(\theta)\sin(k\theta)}{B(\theta)\cos(k\theta)} = -\tan(k\theta)$$

$\therefore \tan\phi = \tan(-k\theta) \Rightarrow \phi(\theta) = -k\theta$

- The phase function $\phi(\theta)$ includes linear phase term and also accommodates for the sign changes in $B(\theta)$.
- Since -1 can be expressed as $e^{\pm j\pi}$, phase jumps of $\pm\pi$ will occur at frequencies where $B(\theta)$ changes sign. Therefore,

If $B(\theta) > 0$, then $\phi(\theta) = -k\theta$.
If $B(\theta) < 0$, then $\phi(\theta) = -k\theta \pm \pi$..

Phase Jumps

- A jump of $\pm 2\pi$ occurs to maintain the phase function within the principal value range of [$-\pi$ and π]
- A jump of $\pm\pi$ occurs when $B(\theta)$ undergoes a change of sign
- The sign of the phase jump is chosen such that the resulting phase function is odd and, after the jump, lies in the range [$-\pi$ and π]

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 10: Low-pass FIR filter

- Calculate the magnitude and phase responses of the 3-sample averager given by;

$$h[n] = \begin{cases} \frac{1}{3} & -1 \le n \le 1 \\ 0 & otherwise \end{cases}$$

Answer:

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n} = \sum_{n=-1}^{1} h(n)z^{-n} = \frac{1}{3}z^{-1} + \frac{1}{3}z^{0} + \frac{1}{3}z^{1}$$

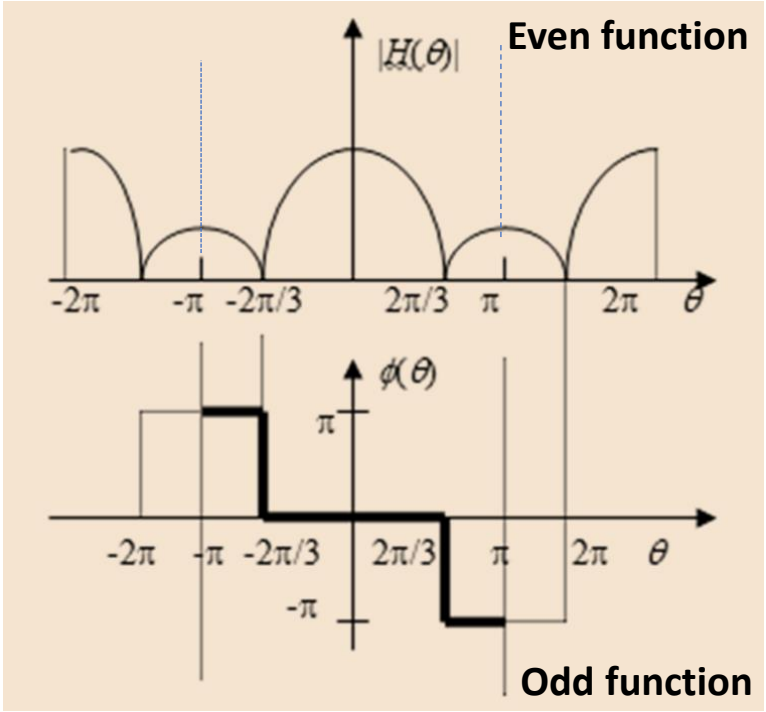
$$H(z) = \frac{1}{3}[z^{-1} + 1 + z^{1}]$$

$$H(\theta) = H(z)\Big|_{z=e^{j\theta}} = \frac{1}{3}\left[e^{-j\theta} + 1 + e^{j\theta}\right] = \left[1 + e^{j\theta} + e^{-j\theta}\right] \cdot \frac{1}{3}$$

$$H(\theta) = \frac{1}{3}[1 + 2\cos\theta] \Rightarrow |H(\theta)| = \left|\frac{1}{3}[1 + 2\cos\theta]\right|$$

$$\begin{cases} \emptyset(\theta) = 0 & H(\theta) \ge 0 \quad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \emptyset(\theta) = 0 \pm \pi & H(\theta) < 0 \quad -\pi \le \theta \le -\frac{2\pi}{3} \ \& \ \frac{2\pi}{3} < \theta < \pi \end{cases}$$

- The appropriate sign of π must be chosen to make $\phi(\theta)$ an odd function of frequency.



Digital Filter Fundamentals: Magnitude and Phase Responses

Example 11: Low-pass FIR filter

- Calculate the magnitude and phase responses of the 3-sample averager given by;

$$h[n] = \begin{cases} \frac{1}{3} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Answer:

$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2} \Rightarrow H(\theta) = \frac{1}{3} + \frac{1}{3}e^{-j\theta} + \frac{1}{3}e^{-j2\theta}$$

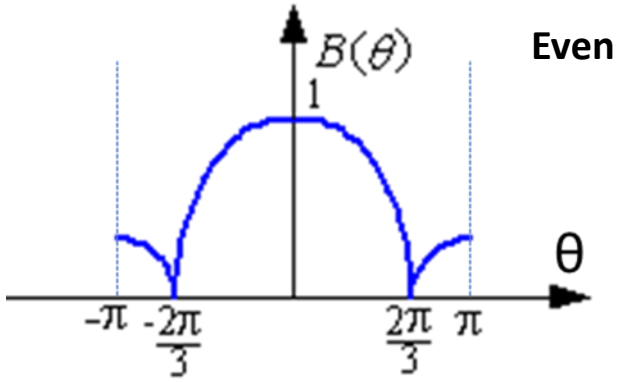
$$H(\theta) = \frac{1}{3}e^{-j\theta}\left[e^{j\theta} + 1 + e^{-j\theta}\right]$$

$$\therefore H(\theta) = e^{-j\theta}\underbrace{\frac{1}{3}[1 + 2\cos\theta]}_{B(\theta)}$$

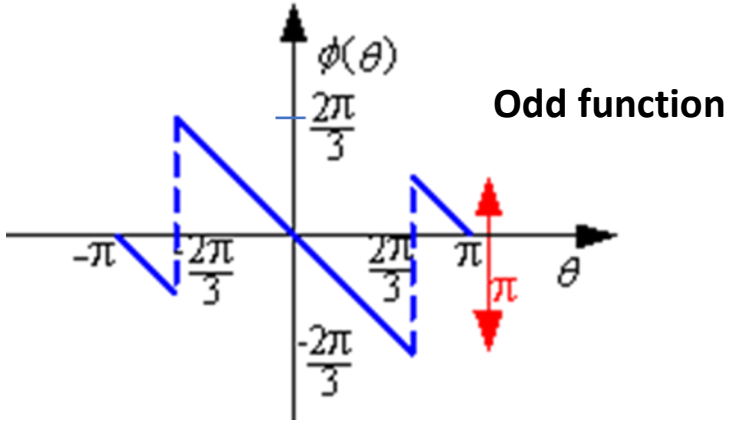
$$|H(\theta)| = |B(\theta)| = \frac{1}{3}[1 + 2\cos\theta] \quad \text{and} \quad \emptyset(\theta) = -\theta;$$

$$\begin{cases} \emptyset(\theta) = -\theta; & B(\theta) \ge 0 \quad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \emptyset(\theta) = -\theta \pm \pi & B(\theta) < 0 \quad -\pi \le \theta \le -\frac{2\pi}{3} \ \& \ \frac{2\pi}{3} < \theta < \pi \end{cases}$$

- Phase is undefined at points $|H(\theta)| = 0$ or B(θ)=0.



Even function



Odd function

Phase Jumps: A jump of ± π occurs when $B(\theta)$ undergoes a change of sign.

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 12:

- Find the magnitude and phase response of the following:

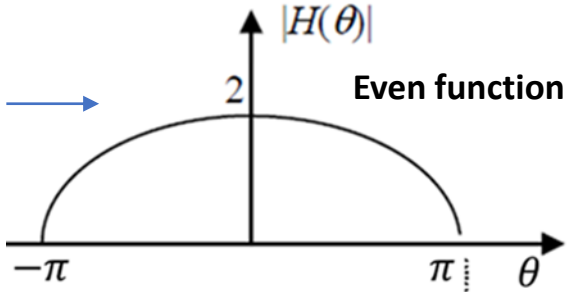
$h(0) = \frac{1}{2}, \quad h(1) = 1, \quad h(2) = \frac{1}{2}, \quad h(n) = 0$, otherwise.

Answer:

$$H(z) = \frac{1}{2}z^0 + 1z^{-1} + \frac{1}{2}z^{-2} = \frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2}$$

$$H(\theta) = \frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta} = e^{-j\theta}\left[\frac{1}{2}e^{j\theta} + 1 + \frac{1}{2}e^{-j\theta}\right]$$

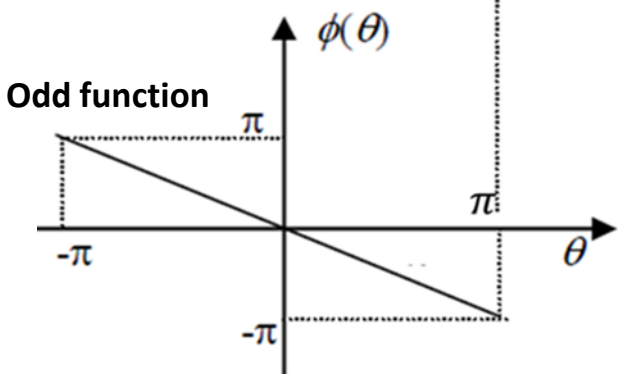
$$H(\theta) = e^{-j\theta}\underbrace{[1 + cos\,\theta]}_{B(\theta)};$$



B(θ)is never negative; therefore, there is no phase jumps of $\pm\pi$.

Phase Response:

$\emptyset(\theta) = -\theta$

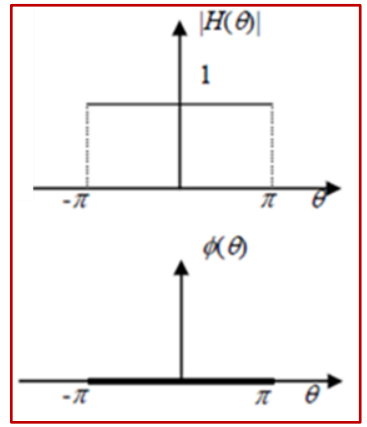


Example 13:

- **Case 1:**

$$\left.\begin{array}{ll}\delta(n) = 1 & n = 0 \\ \delta(n) = 0 & otherwise\end{array}\right\}$$

$$\therefore\ H(\theta) = 1\ ;$$
$$\emptyset(\theta) = 0$$



- **Case 2:** $h[n] = \delta[n-k];$

$$\therefore H[z] = 1\,z^{-k};$$

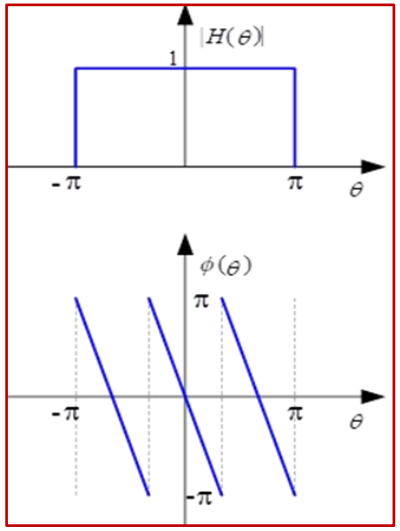
$$H(\theta) = 1\,e^{-jk\theta}$$

$$\mathrm{H}(\theta) = \mathrm{B}(\theta) = 1\ ;$$

$$\phi(\theta) = -\theta\,k;$$

Assume k =3 ; $\emptyset(\theta) - 3\theta$

When phase exceeds ±π range a jump of ±2π is needed to bring the phase back into ±π range.



Digital Filter Fundamentals: Magnitude and Phase Responses

Example 14: Magnitude and Phase Responses of a Low-pass IIR filter

- Determine and sketch the magnitude and phase response of 1st order recursive filter (IIR filter):

$$y[n] = x[n] + ay[n-1] \qquad 0 < a < 1$$

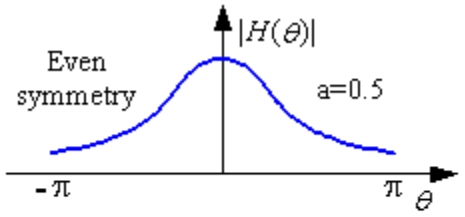
Answer:

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1}{1-az^{-1}}; \qquad H(\theta) = \frac{1}{1-ae^{-j\theta}}$$

$$H(\theta) = \frac{1}{1-ae^{-j\theta}} \cdot \frac{1-ae^{j\theta}}{1-ae^{j\theta}} = \frac{1-a\cos\theta}{1-2a\cos\theta+a^2} - \frac{ja\sin\theta}{1-2a\cos\theta+a^2}$$

$$\tan\emptyset = \frac{-\dfrac{a\sin\theta}{1-2a\cos\theta+a^2}}{\dfrac{1-a\cos\theta}{1-2a\cos\theta+a^2}} = \frac{-a\sin\theta}{1-a\cos\theta}$$

Phase: $\emptyset(\theta) = \tan^{-1}\left[\frac{-a\sin\theta}{1-a\cos\theta}\right]$;

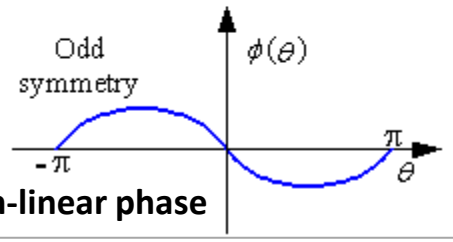


Magnitude:

$$|H(\theta)|^2 = H(\theta).H^*(\theta);$$

$$= \frac{1}{1-ae^{-j\theta}} \cdot \frac{1}{1-ae^{j\theta}}$$

$$|H(\theta)| = \frac{1}{1-2a\cos\theta+a^2};$$



Non-linear phase

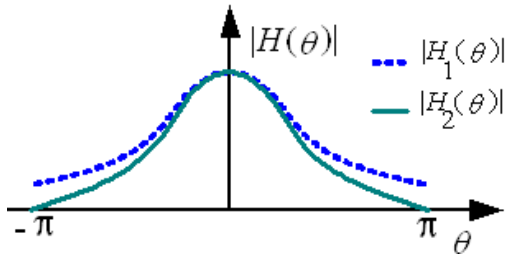
Example 15: Magnitude Responses of IIR filters

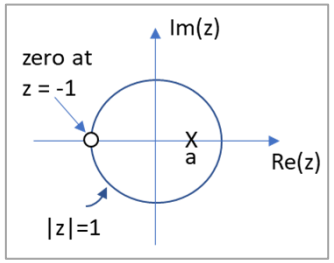
(a) $H_1(z) = \frac{k_0}{1-az^{-1}} \Rightarrow \frac{1-a}{1-az^{-1}} \qquad 0 < a < 1$

The gain k_0 can be selected as $1 - a$, so that the **low-pass filter** has unity gain at $\theta = 0$.

(b) $H_2(z) = k_0\frac{1+z^{-1}}{1-az^{-1}}$; In this case, $k_0 = \frac{1-a}{2}$ (for unity gain at $\theta = 0$).

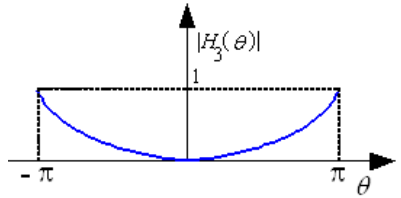
The addition The addition of a zero at z = -1 further attenuates the response of the filter at high frequencies (see magnitude response below)

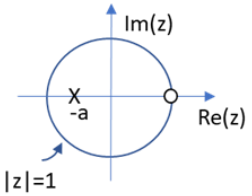




(c) We can obtain simple **high-pass filters** by reflecting (folding) the pole-zero locations of the lowpass filters about the imaginary axis in the z-plane

$$H_3(z) = \frac{1-a}{2} \cdot \frac{1-z^{-1}}{1+az^{-1}}$$





Digital Filter Fundamentals: Magnitude Responses

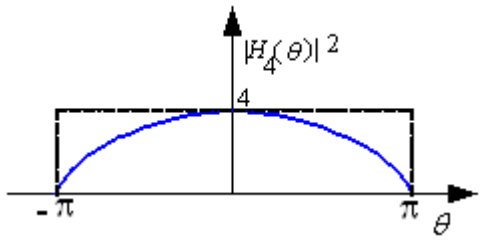
Example 16: FIR filters

- **Low-pass filter:**

$H_4(z) = 1 + z^{-1}$;

$y[n] = x[n] + x[n-1]$

$|H_4(\theta)|^2 = 2(1+\cos\theta)$

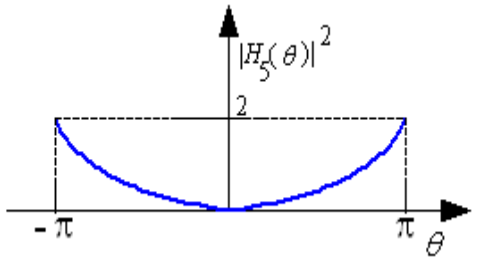


- **High-pass filter:**

$H_4(z) = 1 - z^{-1}$;

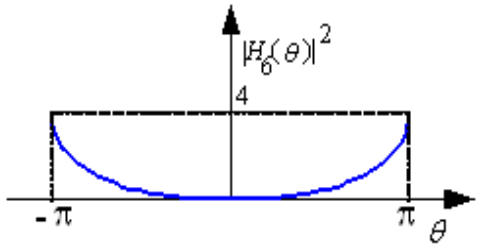
$y[n] = x[n] - x[n-1]$

$|H_5(\theta)|^2 = 2(1-\cos\theta)$



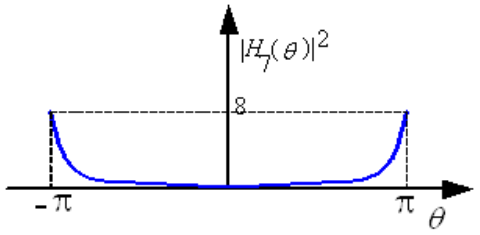
- **High-pass filter:**

$H_6(z) = (1 - z^{-1})^2$



- **High-pass filter:**

$H_7(z) = (1 - z^{-1})^3$

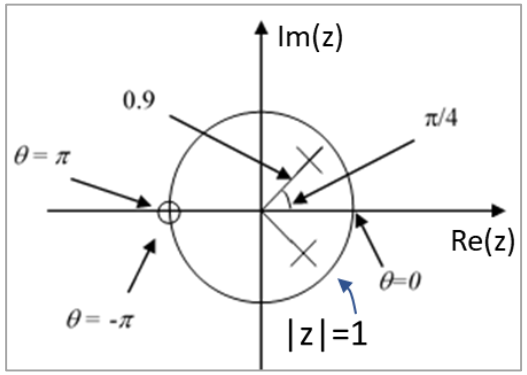


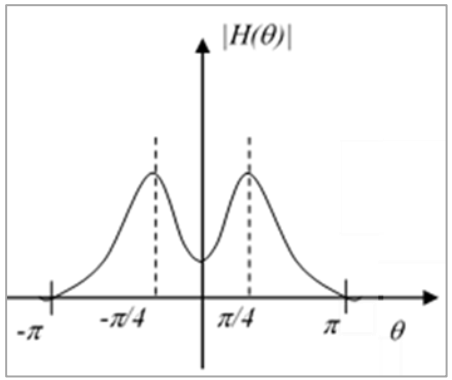
Example 17: IIR Resonant filters

- Sketch an approximate magnitude response for the system having the transfer function

$$H(z) = \frac{1 + z^{-1}}{(1 - 0.9e^{j\frac{\pi}{4}}z^{-1})(1 - 0.9e^{-j\frac{\pi}{4}}z^{-1})}$$

- The system has a zero at z = -1 & poles at $z = 0.9\, e^{-j\frac{\pi}{4}}$
- Magnitude response will be zero at $\theta = \pi$ and large at $\theta_0 = \pm\, \pi/4$ because the poles are close to the unit circle.





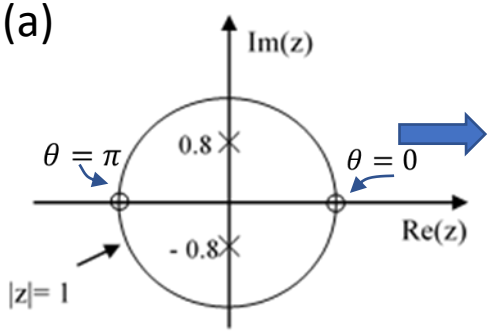
- A complex pole at θ_0 causes a peak in the magnitude response at that frequency.
- A zero on the unit circle will force the magnitude response $|H(\theta)|$ at that frequency to zero.

Digital Filter Fundamentals: Magnitude Responses

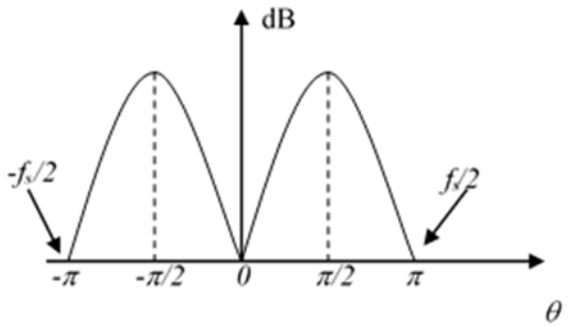
Example 18: IIR resonant filters

- Sketch an approximate magnitude response for each of the pole-zero maps given below:

(a)



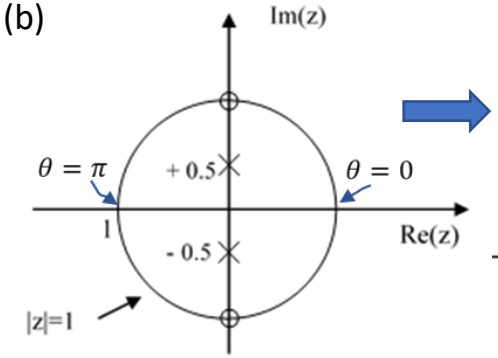
Pole-zero map



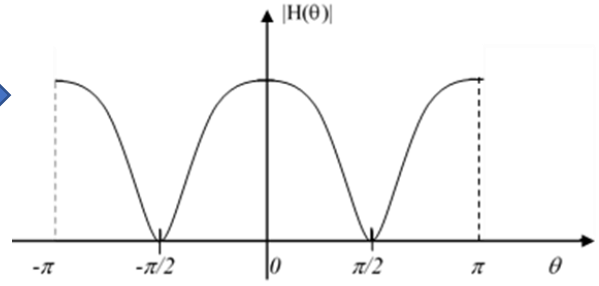
Magnitude response

A complex pole at $\theta_0 = \frac{\pi}{2}$ causes a peak in the magnitude response at θ_0 . Zeros on the unit circle at $\theta = 0$ and $\theta = \pi$ will force the magnitude response at those frequencies to zero.

(b)



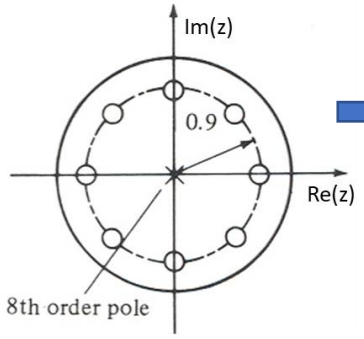
Pole-zero map

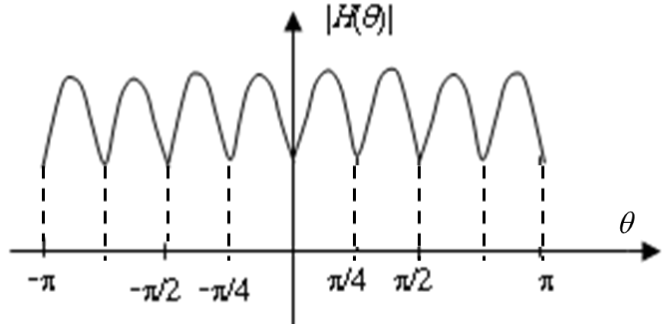


Magnitude response

Example 19: FIR resonant filters

- Sketch roughly the magnitude response corresponding to the pole-zero pattern given below:

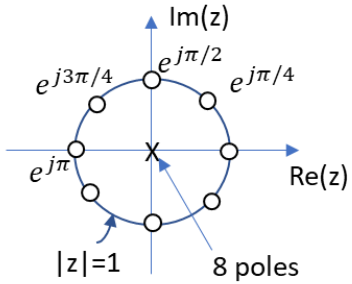


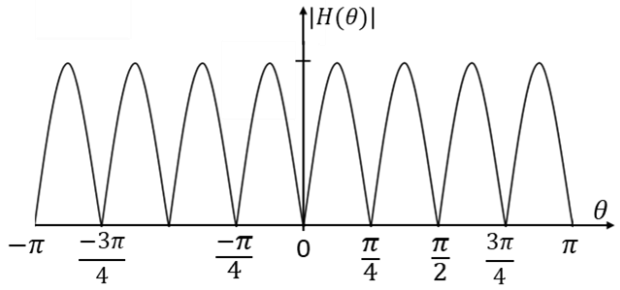


Magnitude response

Note: Zeroes near the unit circle cause a dip in the magnitude response.

- Sketch roughly the magnitude response corresponding to the pole-zero pattern (there *N =8* equally spaced zeros over the unit circle with the first zero at *z* = 0) given below:





ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (pp. 3-10)
- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
- Minimum, Maximum, Mixed Phase Filters (PP. 24-28)
- All-Pass filters (pp. 29-31)
- Tutorial questions (pp. 32-38)

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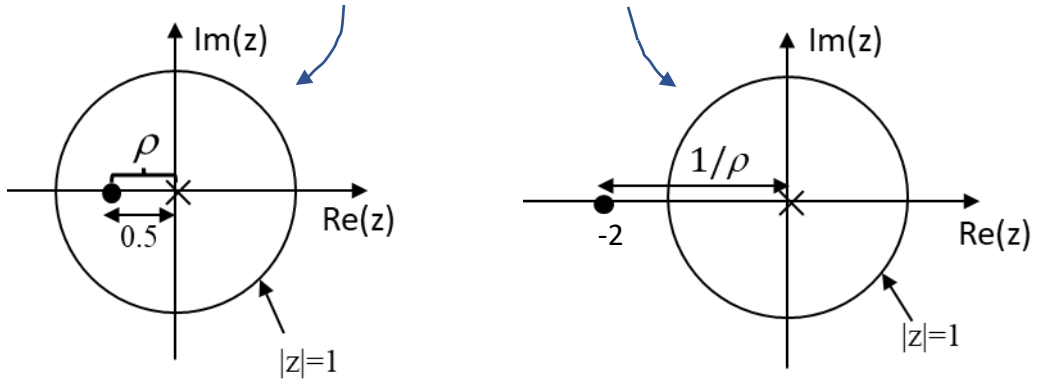
Term 3, 2021

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Filters

FIR filter: Reciprocal roots

- The magnitude characteristics of two FIR filters $H_1(z)$ and $H_2(z)$ are **identical** , because the zeros of the FIR filters are reciprocals.
- If we **reflect** a zero with magnitude $|z| = \rho$, that is inside the unit circle into a zero with magnitude $|z| = 1/\rho$ outside the unit circle the magnitude characteristic of the system is unaltered, but the phase response changes.
- Let us consider two FIR filters of transfer functions :

$$H_1(z) = 1 + \frac{1}{2}z^{-1}; \qquad H_2(z) = \frac{1}{2} + z^{-1}$$

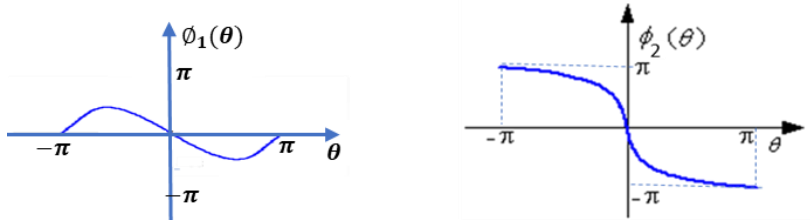


- $H_2(z)$ is the reverse of the system $H_1(z)$. This is due to the reciprocal relationship between the zeros of $H_1(z)$ and $H_2(z)$.

$$H_1(\theta) = 1 + (1/2)\, e^{-j\theta}; \qquad H_2(\theta) = 1/2 + e^{-j\theta}$$

$$|H_1(\theta)| = |H_2(\theta)| = \sqrt{5/4 + \cos\theta}$$

- Phase Response:



- We observe that the phase characters $\phi_1(\theta)$ begins at zero phase at frequency $\theta = 0$ and terminates at zero phase at the frequency $\omega = \pi$. Hence the net phase change. This filter is called **minimum phase** filter

$$\emptyset_1(\pi) - \emptyset_1(0) = 0$$

- On the other hand, the phase characteristic for the filter with the zero outside the unit circle undergoes a net phase change. This filter is called **maximum phase** filter.

$$\emptyset_2(\pi) - \emptyset_2(0) = -\pi$$

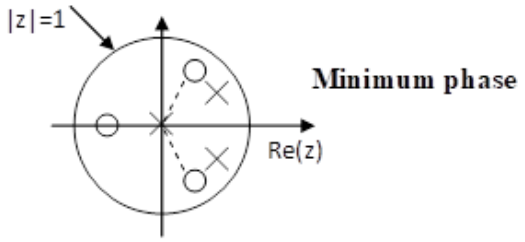
- If a filter with M zeros has some of its zeros inside the unit circle and the remaining outside the unit circle, it is called a **mixed-phase** system.

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase System

Minimum, Maximum & Mixed Phase Filters

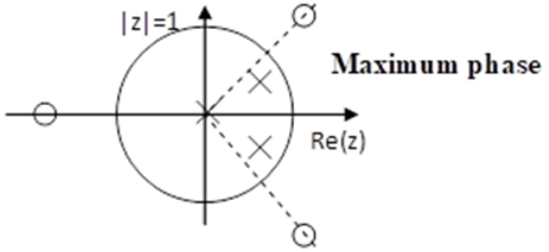
- A minimum-phase property of FIR filter carries over to IIR filter. Let us consider:

$$H(z) = \frac{B(z)}{A(z)}$$

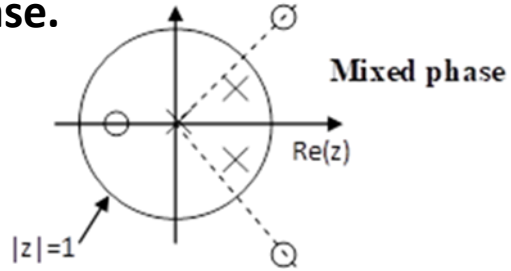


is called **minimum phase** if all its poles and zeros are inside the unit circle.

- If all the zeros lie outside the unit circle, the system is called **maximum phase**.



- If zeros lie both inside and outside the unit circle, the system is called **mixed-phase.**



Note:

- For a given magnitude response, the minimum-phase system is the causal system that has the smallest phase at every frequency (θ).

- Since the derivative of the phase characteristic of the filter (group delay) is a measure of the time delay that signal frequency components undergo in passing through the filter, a minimum phase characteristic implies a minimum delay function, while a maximum-phase characteristic implies that the delay characteristic is also maximum

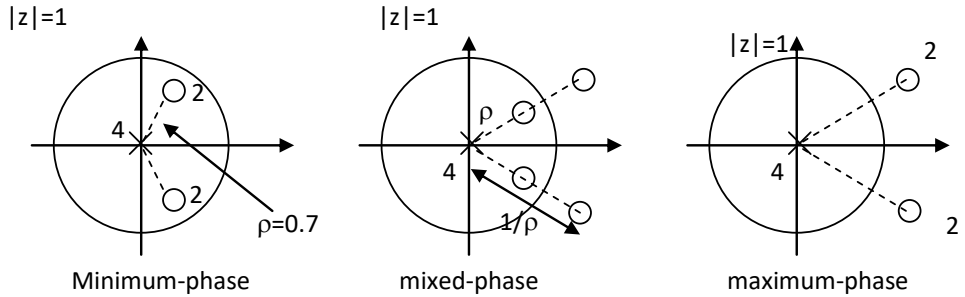
Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase System

Example 20:

- Consider a fourth-order all-zero filter containing a double complex conjugate set of zeros located at

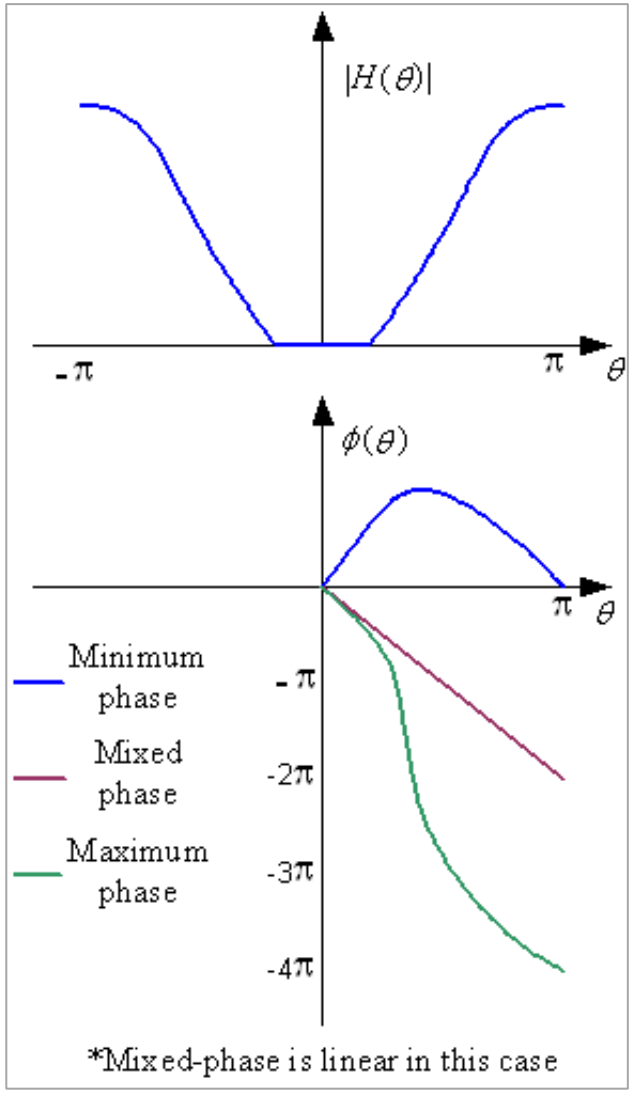
$$z = 0.7e^{\pm j\frac{\pi}{4}}$$

- The minimum-phase, mixed phase and maximum phase system pole-zero patterns having identical magnitude response are shown below:



- The magnitude response and the phase response of the three systems are shown on the right.

- The minimum-phase system seems to have the phase with the smallest deviation from zero at each frequency.



Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase System

Example 21:

- A third order FIR filter has a transfer function $G(z)$ given by:

$$G(z) = (6 - z^{-1} - 12z^{-2})(2 + 5z^{-1})$$

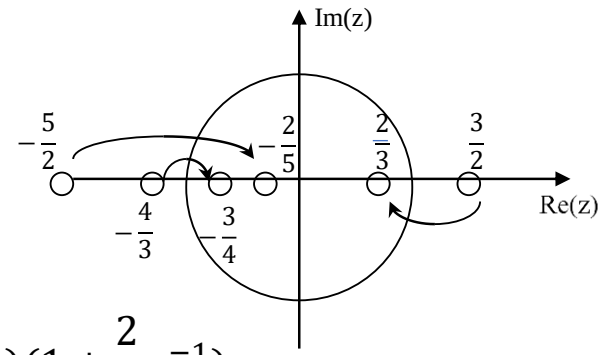
From $G(z)$, determine the transfer function of an FIR filter whose magnitude response is identical to that of $G(z)$ and has a minimum phase response.

Answer: First Factorise G(z):

$$G(z) = (2 - 3z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

$$G(z) = 12(1 - \frac{3}{2}z^{-1})(1 + \frac{4}{3}z^{-1})(1 + \frac{5}{2}z^{-1})$$

- All the zeros of G(z) are outside the unit circle; So, reflect all the zeros inside the unit circle to obtain the minimum-phase filter.

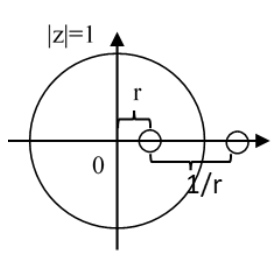


The minimum phase filter:

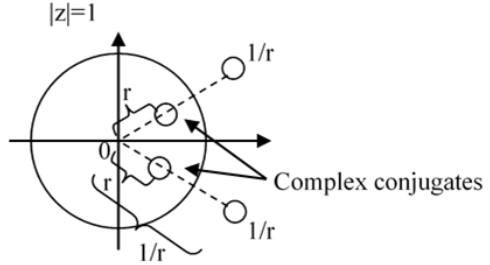
$$P(z) = 60(1 - \frac{2}{3}z^{-1})(1 + \frac{3}{4}z^{-1})(1 + \frac{2}{5}z^{-1})$$

Example 22: Reciprocal pairs of zeros for linear phase filters

- The zeros of a linear-phase filter must occur in reciprocal pairs and exhibit conjugate reciprocal symmetry as shown below.

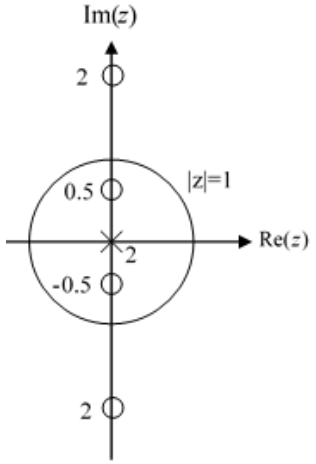


For a real zero



For a complex zero

- The pole-zero plot of a filter H(z) is given below. Is $H(z)$ a linear phase filter?



The four zeros at $z = j0.5$, $z = j2$, $z = -j0.5$ and $z = -j2$ are conjugate reciprocal symmetry. All poles are at $z = 0$.

Therefore, this is a linear phase filter.

$H(z) = z^2 + 4.25 + z^{-2}$

$\therefore h[n] = \{1, 0, 4.25, 0, 1\}$ <– even symmetry about n=0.

ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (pp. 3-10)
- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
- Minimum, Maximum, Mixed Phase Filters (PP. 24-28)
- All-Pass filters (pp. 29-31)
- Tutorial questions (pp. 32-38)

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Term 3, 2021

Digital Filter Fundamentals: All-Pass filters

All-Pass Filters

- An all-pass filter is one whose magnitude response is constant for all frequencies, but whose phase response is not identically zero.
- The simplest example of an all-pass filter is a pure delay system with system function $H(z) = z^{-k}$
- A more interesting all-pass filter is one that is described by

$$H(z) = \frac{a_L + a_{L-1}z^{-1} + \cdots + a_1 z^{-L+1} + a_0 z^{-L}}{1 + a_1 z^{-1} + \cdots + a_L z^{-L}}$$

where $a_0 = 1$ and all coefficients are real.

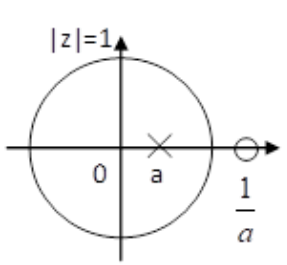
- If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{L} a_k z^{-k} \qquad a_0 = 1$$

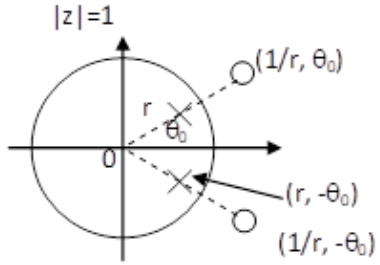
$$\therefore H(z) = z^{-L}\frac{A(z^{-1})}{A(z)}$$

- It can be shown that $|H(\theta)|^2 = H(z)\cdot H(z^{-1})|_{z=e^{j\theta}} = 1$ which is an all-pass filter.

- Furthermore, if $|z_0|$ is the modulus of a pole of $H(z)$, then $1/|z_0|$ is the modulus of a zero of $H(z)$ {i.e. the modulus of poles and zeros are reciprocals of one another}.
- The figure shown below illustrates typical pole-zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter.



All-pass



All pass filter

$$H(z) = \frac{1-\frac{1}{a}z^{-1}}{1-az^{-1}} \qquad |a| < 1 \text{ for stability}$$

- We can easily show that the magnitude response is constant.

$$|H(\theta)|^2 = |H(\theta)\cdot H^*(\theta) = H(z)\cdot H(z^{-1})|_{z=e^{-j\theta}}$$

$$= \frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-\frac{1}{a}e^{j\theta}}{1-ae^{j\theta}} = \frac{1-\frac{2}{a}\cos\theta+\frac{1}{a^2}}{1-2a\cos\theta+a^2} = 1/a^2$$

Digital Filter Fundamentals: All-Pass filters

Example 23:

- Show that the following transfer function $H_1(z)$ can be obtained using a parallel connection of a constant and an all-pass filter.

$$H_1(z) = \frac{10 + 6z^{-1}}{3 + z^{-1}}$$

Answer:

$$H_1(z) = \frac{9 + 3z^{-1} + 1 + 3z^{-1}}{3 + z^{-1}}$$

$$= 3 + \frac{1 + 3z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \underbrace{3}_{constant} + \underbrace{\frac{1 + 3z^{-1}}{3 + z^{-1}}}_{All-passfilter}$$

Example 24: All-pass filter

- Consider the filter described by

$H(z) = \frac{a+cz^2}{c+az^2}$ where a & c are constants.

Show that the magnitude response $|H(\theta)|$ is unity for all θ.

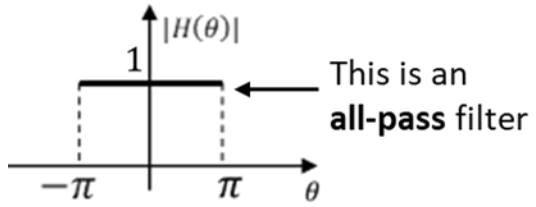
Answer:

$$H(z) = \frac{c+az^{-2}}{a+cz^{-2}}; \qquad H(\theta) = \frac{c+ae^{-j2\theta}}{a+ce^{-j2\theta}}$$

$$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta)$$

$$= \frac{c + ae^{-j2\theta}}{a + ce^{-j2\theta}} \cdot \frac{c + ae^{j2\theta}}{a + ce^{j2\theta}}$$

$$= \frac{c^2 + a^2 + ac[e^{j2\theta} + e^{-j2\theta}]}{a^2 + c^2 + ac[e^{j2\theta} + e^{-j2\theta}]} = 1$$



ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

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Term 3, 2021

Topic 3: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

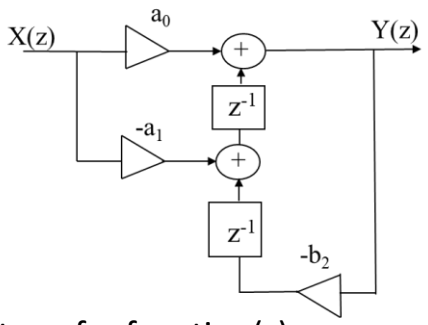
1. A digital filter structure is shown below. Calculate the dc gain of the filter, in terms of the filter coefficients. (Tick the correct answer)

   (a) $1 + a_0 - a_1$

   (b) $\frac{a_0 - a_1}{1 + b_2}$

   (c) $\frac{a_0 - a_1}{1 - b_2}$

   (d) Non of the above



2. Tick which of the following transfer function(s) represent high pass filters, where $0 < \alpha < 1$.

   $(a)\ \ H(z) = 1 - \alpha\, z^{-1}$ $\quad (b)\ \ H(z) = \frac{1}{1 - \alpha\, z^{-1}}$

   $(c)\ \ H(z) = 1 + \alpha\, z^{-1}$ $\quad (d)\ \ H(z) = \frac{1}{1 + \alpha\, z^{-1}}$

3. The Digital Filter $H(z) = \frac{2 + 2z^{-1}}{3 - z^{-1}}$; (Tick the correct answer)

   a) is a low-pass filter
   b) is a high-pass filter
   c) is an all-pass filter

4. A maximum phase filter has: (Tick the correct answer)

   a) All zeros and poles outside the unit circles
   b) All poles outside the unit circle
   c) All zeros outside the unit circle
   d) All poles and zeros inside the unit circle

SOLO – Level 1

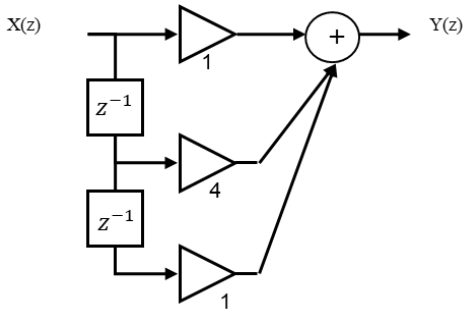
5. Which of the following is/are linear phase filters: (Tick the correct answer)

   (a) $H(z) = 0.25 + 0.5\, z^{-1} + 0.25\, z^{-2}$ (b) $H(z) = z^{-4}$

   (c) $H(z) = 1 - z^{-2}$ (d) $H(z) = 1 + 2z^{-1} + 2\, z^{-2} - z^{-3}$

6. What can be said of the following filter ? (Tick the correct answer)

   a) It has minimum phase and a finite impulse response.
   b) It has linear phase and an infinite impulse response.
   c) It has an infinite impulse response.
   d) It has linear phase and a finite impulse response



7. Which of the following system is linear, time invariant, causal and stable? (Tick the correct answer)

   a) y[n] = |x[n]|
   b) y[n] = x[n] + n x[n+1]
   c) y[n] = x[-n]
   d) $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$

Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

1. Compute the 3dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}}; \quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}, \qquad 0 < a < 1$$

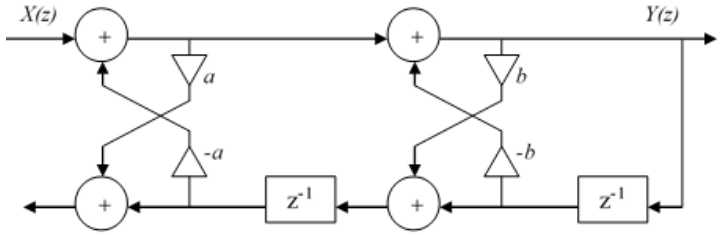
   Which filter has a smaller 3dB bandwidth?

2. Find the magnitude response of the filter *H(z)* and comment on your result.

$$H(z) = \frac{1+3z^2}{3+z^2}$$

   Draw the canonic realization of the filter *H(z)*.

3. Determine the frequency response $H(\theta)$ of the following filter:



4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1-b_1z^{-1})^2}$$

   Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

5. Implement a parallel structure for the following *H(z):*

$$H(z) = \frac{z^2-4z+3}{z(z+0.5)^2}$$

SOLO – Level 2

6. A notch filter is given by $H(z) = \frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta = 0, \frac{\pi}{2}, and\ \pi$. Sketch the magnitude response in the interval $-\pi \le \theta \le \pi$.

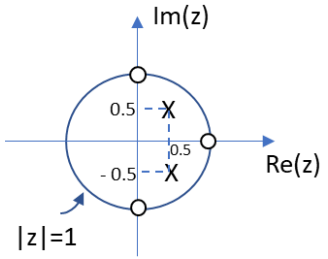
7. Determine and sketch the magnitude and phase response of the following filters

   (a) $y[n] = \frac{1}{2}(x[n] - x[n-1])$ (b) $y[n] = x[n] - x[n-8]$

   (c) $y[n] = x[n-4]$

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below:

   (b) Using the pole-zero map , sketch an approximate magnitude response of the H(z) .



9. A Finite impulse filter filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$.Determine b_4 to b_6 such t hat this filter has linear phase characteristics.

10. Implement the following transfer function in cascade structure , in direct form II and in parallel structure. All coefficients are real.

$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

1. Show that both digital filters given below have the same magnitude response:

   (a) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-i]$ (b) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-m-i]$

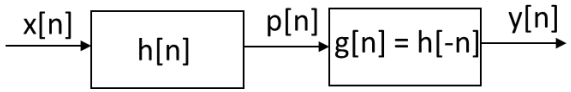
   where $y[n]$ is the output, $x[n]$ is the input and c_i are the coefficients.

2. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2+z+0.5)}$$

   Is *H(z)* a stable function? If it is not stable, find a stable transfer function *G(z)* such that $|G(\theta)| = |H(\theta)|$. Is there any transfer function having the same magnitude response as *H(z)*?

3. Show that if *x*[n] is a real valued signal, the structure shown below has a zero-phase response; where *h*[n] is the impulse response of the filter H(z) given by; $H(z) = (\frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2})$.



4. The difference equation of a digital filter is given by

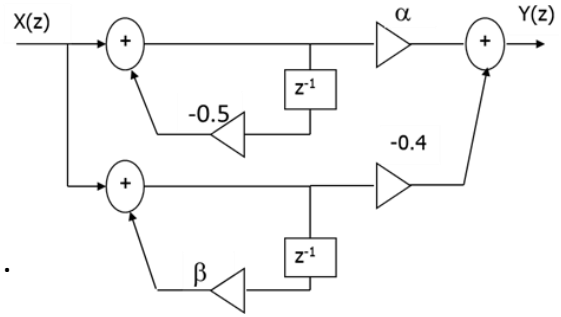
$$y[n] = (1/8)\{x[n] - x(n-8)\} - y(n-2)$$

   (a) Find the transfer function for the above filter.
   (b) Using the transfer function obtained in part (a) above, draw the pole zero map for the filter and sketch the corresponding magnitude response.

SOLO – Level 3

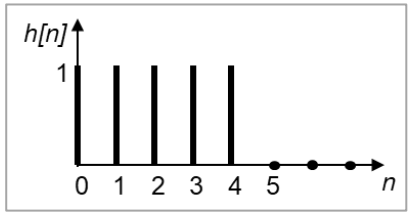
5. The following figure shows an IIR filter structure, and its transfer function given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$$



   Determine the values of the Multiplier Coefficients α and β.

6. A FIR filter has an impulse response as shown below:

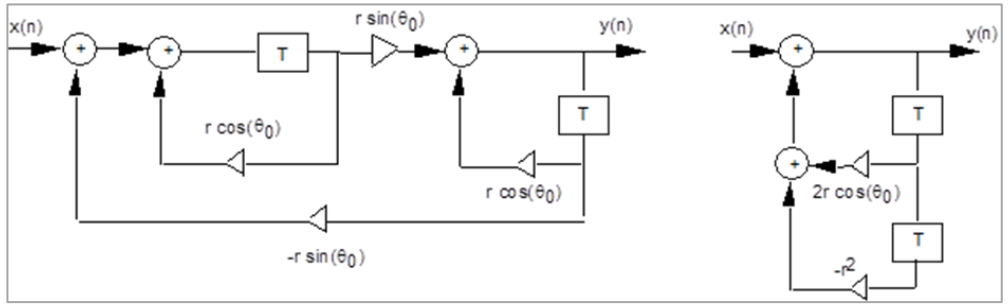


   (a) Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$

   (b) Draw an implementation of this filter as a cascade structure.

7. Show that the systems (i) and (ii) shown below are equivalent:



Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

8. A third order FIR filter has a transfer function G(z) given by

$$G(z) = (9 + 6z^{-1} - 8z^{-2})(2 + 5z^{-1}).$$

From G(z), determine the transfer function of a FIR filter whose magnitude response is identical to that of G(z) and has

(a) a minimum phase response
(b) a maximum phase response

9. Analyse the digital structure shown on the right:

(a) Determine its transfer function $H(z)$ =(Y(z))/(X(z))

(b) What should be the value of the multiplier coefficient K so that the transfer function *H(z)* that you obtained in (a) above has a unity gain at $\theta = 0$?

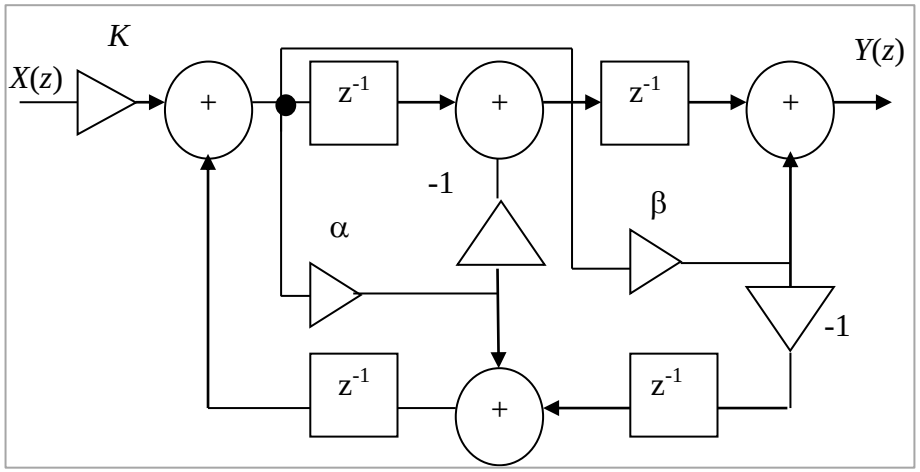
(c) What should be the value of the multiplier coefficient *K* so that *H*(*z*) has a unity gain at $\theta = \pi$?

(d) Is there a difference between these two values of *K* obtained in Question (b) and (c)? If not, why not?

10. (a) Calculate the magnitude and phase responses of the following FIR filter:

$$y[n] = \frac{1}{4}[x[n] - 2x[n-1] + x[n-2]]$$

(b) Sketch the magnitude and phase responses.
(c) Show that the above FIR filter has a constant group delay.



Topic 3: Tutorial questions (Level 4)

SOLO – Level 4

1. The frequency response of an ideal differentiator is given by

$$H(\theta) = j\theta e^{-j\tau} \text{ - } \pi \leq \theta \leq \pi$$

This response is periodic with period 2π. The quantity τ is the delay of the system in samples.

   (a) Sketch the magnitude and phase responses of the system for $-\pi \leq \theta \leq \pi$.

   (b) Find the impulse response *h[n]* of the system as a sum of a sine and cosine functions.

2. Determine the magnitude and phase response of the multipath channel *y[n] = x[n] + x[n-M]*. At what frequencies $H(\theta) = 0$?

3. Let a causal linear time invariant discrete-time system be characterised by a real impulse response *h*[*n*] with a DTFT of $H(\theta)$. Consider the system shown in Figure A , where *x*[*n*] is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ $\{G(z) = {}^{Y(z)}/_{X(z)}\}$ in terms of $H(\theta)$ and show that it has a zero-phase response.

4. An analogue system may be implemented as shown in Figure B. Using the transformation $\frac{1}{s} \rightarrow k\frac{1+z^{-1}}{1-z^{-1}}$ a discrete time system may be obtained; where k is a constant.

   (a) Draw the digital structure of the system, using the minimum number of multipliers and delay units .

   (b) Write the difference equation that governs the behaviour of the system in the form $y[n] = 2\cos[\theta_0]\, y[n-1] - y[n-2]$ and find the value of $\cos\theta_0$.

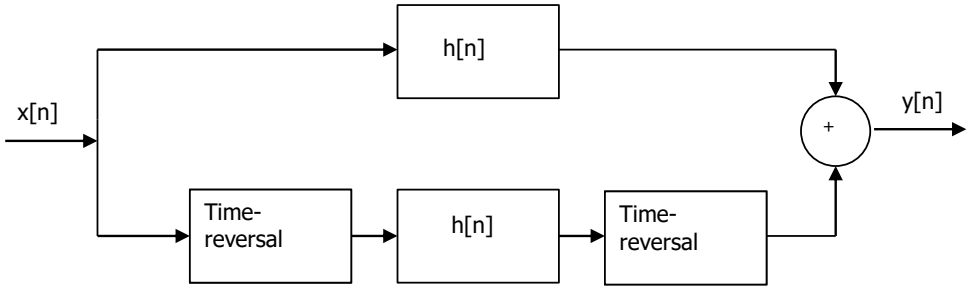


Figure A

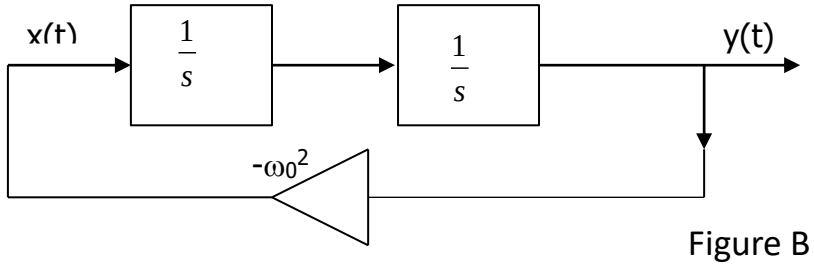


Figure B

Topic 3: Tutorial questions (Level 5)

SOLO – Level 5

1. (a) Find the group delay for each of the following system, where $|\alpha| < 1$

(i) $H_1(\theta) = 1 - \alpha e^{-j\theta}$ (ii) $H_2(\theta) = \frac{1}{1-\alpha e^{-j\theta}}$

(b) using the results obtained in (a) above, show that the group delay of an all-pass filter with a transfer function $H_0(z) = \frac{z^{-1} - \alpha}{1-\alpha z^{-1}}$ is $\frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos(\theta)}$

2. Find the minimum phase system that has a magnitude response given by

$$|H(\theta)|^2 = \frac{\frac{5}{4} - \cos\theta}{\frac{10}{9} - \frac{2}{3}\cos\theta}$$

3. A stable and causal system (H(z)) may always be factored into a product of a minimum phase system with an all-pass system: $H(z) = H_{min}(z) \cdot H_{ap}(z)$ **or** $H(z) = H_{min}(z) \cdot H_{lp}(z)$; where $H_{lp}(z)$ is a linear phase system.

$$H(z) = \frac{(1 - 0.7z^{-1})(1 - j2z^{-1})(1 + j2z^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})}$$

(a) Find a minimum phase system function $H_{min}(z)$ and an all-pass system $H_{ap}(z)$, for the above H(z).

(b) Find a minimum phase system function $H_{min}(z)$ and a linear phase system $H_{lp}(z)$, for the above H(z).

SOLO – Level 5

4. A filter structure with a transfer function $\{H(z) = \frac{Y(z)}{X(z)}\}$ is shown below. You may assume that $N = 3$, $a_0 = 3$ and $H(\theta)|_{\theta=2\pi/3} = 6+j3\sqrt{3}$.

Also $H\left(e^{j\frac{2\pi k}{N}}\right) = a_k$ and $H(e^{-j\frac{2\pi k}{N}}) = a_{-k}$

(a) Determine a_1 & a_{-1} such that this filter has a real valued impulse response *h[n];*

(b) Find H(z) and draw a filter structure using delays elements, multipliers and adders.

(c) Show that this filter structure can be realised as an FIR filter.

